
Test Execution Report

IBA Campus Facility Booking System — Cycle 1 Summary

Prepared by:

Abdul Wasay Imran (27126) Saad Imam (27079)
Muhammad Imad Raza (26953) Syed Muhammad Deebaj Haider Kazmi (26012)

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Contents

1	Executive Summary	3
1.1	Overview	3
1.2	Summary of Test Run Performance	3
1.3	Release Recommendation	3
2	High-Level Execution Metrics	3
3	Detailed Test Case Status	4
3.1	Authentication & Authorisation (TC-AUTH)	4
3.2	Room Booking Request (TC-BOOK)	4
3.3	Booking Management — Program Office (TC-PO)	4
3.4	Booking Cancellation (TC-CANCEL)	6
3.5	Admin System Management (TC-ADMIN)	6
3.6	Data Integrity & Audit (TC-DATA)	6
4	Defect Resolution Summary	7
4.1	Conditional DB Unique Constraint Fix (DEF-002 — Resolved)	7
4.2	Cancellation Status Fix (DEF-001 — Resolved)	7
4.3	DTO Range Validation Fix (DEF-010 — Resolved)	7
4.4	Audit Trail Field Fix (DEF-015 — Resolved)	7

1 Executive Summary

1.1 Overview

This document serves as the official Test Execution Report for the first execution cycle of the IBA Campus Facility Booking System. The test suite was executed against the local testing environment consisting of a React frontend, NestJS backend, and an isolated Supabase PostgreSQL instance. Testing covered three distinct layers: unit testing of core services, integration testing of REST API endpoints, and end-to-end (E2E) automated browser workflows (via Playwright).

1.2 Summary of Test Run Performance

The overall test execution yielded excellent results, demonstrating high stability across all layers of the test suite:

- **Total Test Cases Run:** 124 (39 Unit, 43 Integration, 42 E2E)
- **Passed Tests:** 124 (39 Unit, 43 Integration, 42 E2E)
- **Failed Tests:** 0
- **Overall Pass Rate:** 100.0%

1.3 Release Recommendation

Based on the exit criteria defined in Section 9.2 of the Master Test Plan, the system **meets all automated test quality gates and is recommended for release**. All 124 test cases pass across unit, integration, and E2E levels, achieving a 100% pass rate. Previously identified Severity-1 defects (DEF-001, DEF-002, DEF-003, DEF-004, DEF-007, DEF-010, DEF-015) have been resolved.

2 High-Level Execution Metrics

The table below presents the quantitative pass/fail metrics aggregated by functional module across all test levels.

Table 1: Module-Wise Test Execution Summary

Functional Module	Tests Executed	Passed	Failed	Pass Rate (%)	Status
Authentication & Auth.	28	28	0	100.0%	PASSED
Room Booking Request	16	16	0	100.0%	PASSED
PO Booking Management	13	13	0	100.0%	PASSED
Booking Cancellation	22	22	0	100.0%	PASSED
Admin System Management	44	44	0	100.0%	PASSED
Data Integrity & Audit	1	1	0	100.0%	PASSED
Total Summary	124	124	0	100.0%	ALL PASSED

3 Detailed Test Case Status

This section tracks the execution status of individual test cases mapped out in the Test Plan, highlighting the failures and linking them to logged bugs.

3.1 Authentication & Authorisation (TC-AUTH)

This module verified credential checks, session termination, and role-based access control (RBAC).

TC ID	Test Case Description	Level	Status	Failure/Defect Mapping
TC-AUTH-001	Student Login Flow	All	Pass	None
TC-AUTH-002	PO Login Flow	All	Pass	None
TC-AUTH-003	Bad Credentials	All	Pass	None
TC-AUTH-004	Protected URL Redirect	Int/E2E	Pass	None
TC-AUTH-005	Student RBAC Enforcement	Int	Pass	None
TC-AUTH-006	PO RBAC Enforcement	Int	Pass	None
TC-AUTH-007	Logout & Back Button Block	E2E	Pass	None

3.2 Room Booking Request (TC-BOOK)

This module verified the user workflow of making reservations, enforcing validation rules, and temporal limits.

TC ID	Test Case Description	Level	Status	Failure/Defect Mapping	Map-
TC-BOOK-001	Happy Path Booking Submission	All	Pass	None	
TC-BOOK-002	Duplicate Slot Rejection	Int/E2E	Pass	None	
TC-BOOK-003	Blocked Slot Rejection	Int	Pass	None	
TC-BOOK-004/7	Boundary: Out-of-bounds slots	Int	Pass	None	
TC-BOOK-008	Missing purpose field check	All	Pass	None	
TC-BOOK-009	Concurrency: Overlapping requests	Int	Pass	None	
TC-BOOK-010	Past Date Reservation Block	Int	Pass	None	

3.3 Booking Management — Program Office (TC-PO)

This module validated PO dashboard listing, approval, rejection, and conflict-resolution processes.

TC ID	Test Case Description	Level	Status	Failure/Defect Mapping
TC-PO-001	View pending requests	Int/E2E	Pass	None
TC-PO-002	PO approves request	Int/E2E	Pass	None
TC-PO-003	PO rejects request	Int/E2E	Pass	None

TC-PO-004	Conflict Auto-Rejection	E2E	Pass	None
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3.4 Booking Cancellation (TC-CANCEL)

This module verified the ability to void bookings, verifying that the canceled slots are returned to the pool.

TC ID	Test Case Description	Level	Status	Failure/Defect Mapping
TC-CANCEL-001	Student cancels pending booking	Int/E2E	Pass	None
TC-CANCEL-002	Student cancels approved booking	Int/E2E	Pass	None
TC-CANCEL-003	Cancellation exactly at start time	E2E	Pass	None
TC-CANCEL-004	Cancellation after start time block	E2E	Pass	None
TC-CANCEL-005	PO cancels approved booking	Int/E2E	Pass	None

3.5 Admin System Management (TC-ADMIN)

This module validated building, room, and user CRUD operations as well as administrator blocking.

TC ID	Test Case Description	Level	Status	Failure/Defect Mapping
TC-ADMIN-001	Admin adds a new student	All	Pass	None
TC-ADMIN-002	Duplicate ERP constraint check	All	Pass	None
TC-ADMIN-003	Admin adds a PO member	All	Pass	None
TC-ADMIN-004	Admin adds a building	All	Pass	None
TC-ADMIN-005	Admin adds a room	All	Pass	None
TC-ADMIN-006	Admin blocks/disables a room	All	Pass	None
TC-ADMIN-007	Admin unblocks/re-enables room	All	Pass	None

3.6 Data Integrity & Audit (TC-DATA)

This module focused on ensuring the historical state transitions remain valid inside the database schema.

TC ID	Test Case Description	Level	Status	Failure/Defect Mapping
TC-DATA-001	Maintain accurate audit trail	Int	Pass	None

4 Defect Resolution Summary

During this execution cycle, all previously identified defects were resolved. The following subsections document the root causes and the fixes applied.

4.1 Conditional DB Unique Constraint Fix (DEF-002 — Resolved)

The failures of TC-BOOK-009 (Concurrency) and TC-PO-004 (Conflict Resolution) were caused by a structural database flaw. The unconditional unique constraint in `supabase-schema.sql` was replaced with a conditional partial index targeting only active bookings:

```
CREATE UNIQUE INDEX idx_active_bookings ON bookings (room_id, date, slot_id)
  WHERE (status IN ('pending', 'approved'));
```

This allows multiple students to submit concurrent pending requests for the same slot and permits slot recycling after cancellation or rejection.

4.2 Cancellation Status Fix (DEF-001 — Resolved)

The tests TC-CANCEL-001 and TC-CANCEL-005 were failing because the cancel handler in `bookings.service.ts` was routing to the rejection workflow with a hardcoded `'rejected'` status. The fix corrected the status string to `'cancelled'`:

```
async cancel(id: string, requesterId: string, requesterRole: string) {
  ...
  return this.updateStatus(id, 'cancelled', requesterId);
}
```

Cancellations now correctly set the status to `cancelled`, preserving the distinction between rejections and cancellations in the audit log.

4.3 DTO Range Validation Fix (DEF-010 — Resolved)

Test case TC-BOOK-004/7 was failing because the API accepted out-of-bounds slot IDs (such as `slot_id: 0` or `slot_id: 8`), causing an unhandled 500 Internal Server Error. The `CreateBookingDto` was updated with `@Min(1)` and `@Max(7)` decorators, so invalid values are now rejected with a clean 400 Bad Request validation response before reaching the database.

4.4 Audit Trail Field Fix (DEF-015 — Resolved)

During TC-DATA-001, the integration runner was fetching booking records and asserting on the `reviewed_by` field. The fix added `reviewed_by` to the SQL SELECT projection string in `bookings.service.ts`, ensuring the reviewer's identity is returned in the API response and displayed correctly on the frontend audit trail.